Deep Reinforcement Learning

Bonus Unit 2: Automatic Hyperparameter Tuning with Optuna

Hugging Face Deep RL Course
Complete MDX-to-LaTeX study edition

Source: Hugging Face Deep RL Course (original English text).

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1 Introduction

One of the most critical tasks in Deep Reinforcement Learning is to **find a good set of training hyperparameters**.



Figure 1: Optuna library logo. [original image]

Optuna is a library that helps you to automate the search. In this Unit, we'll study a **little bit of the theory behind automatic hyperparameter tuning**. We'll first try to optimize the parameters of the DQN studied in the last unit manually. We'll then **learn how to automate the search using Optuna**.

2 Optuna Tutorial

The content below comes from Antonin's Raffin ICRA 2022 presentations, he's one of the founders of Stable-Baselines and RL-Baselines3-Zoo.

2.1 The theory behind Hyperparameter tuning

Video lecture: The theory behind Hyperparameter tuning
Watch on YouTube (AidFTOdGNFQ).

2.2 Optuna Tutorial

Video tutorial: Optuna
Watch on YouTube (ihP7E76KGOI).

The notebook → here.

3 Hands-on

Now that you've learned to use Optuna, here are some ideas to apply what you've learned:

1. **Beat your LunarLander-v2 agent results**, by using Optuna to find a better set of hyperparameters. You can also try with another environment, such as MountainCar-v0 and CartPole-v1.

2. **Beat your SpaceInvaders agent results**.

By doing this, you'll see how valuable and powerful Optuna can be in training better agents.

Have fun!

Finally, we would love **to hear what you think of the course and how we can improve it**. If you have some feedback then please → fill out this form.

Keep Learning, stay awesome